

InsightStream

Model Card

Following the Mitchell et al. (2018) Framework for Model Reporting

Base Model: Gemini 2.5 Flash | Application: Qualitative Research Synthesis

Document Version: 1.0 | Date: Feb 2025

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1. Model Details

Property	Value
Model Name	Gemini 2.5 Flash (gemini-2.5-flash)
Developer	Google DeepMind
Architecture	Sparse Mixture-of-Experts (MoE)
Model Type	Multimodal large language model (text, image, audio, video input; text output)
Context Window	1,048,576 tokens (1M)
Max Output Tokens	65,536 tokens
Training Data Cutoff	January 2025
Hybrid Reasoning	Supports "Thinking Process" (Chain-of-Thought) visibility. Disabled in InsightStream (temperature=0, structured output mode).
Application Version	InsightStream v2.0 (LangGraph 6-node pipeline)
Application Developer	InsightStream Team
Release Date	June 2025
License	Proprietary (application layer). Gemini API usage governed by Google's Terms of Service.

1.1 Application Architecture

InsightStream wraps Gemini 2.5 Flash inside a 6-node LangGraph StateGraph. The base model is called directly via the langchain-google-genai SDK with structured output (Pydantic schema coercion). No fine-tuning has been applied. All behavioral constraints are imposed through prompt engineering, structured output schemas, and a post-hoc semantic entailment verification layer.

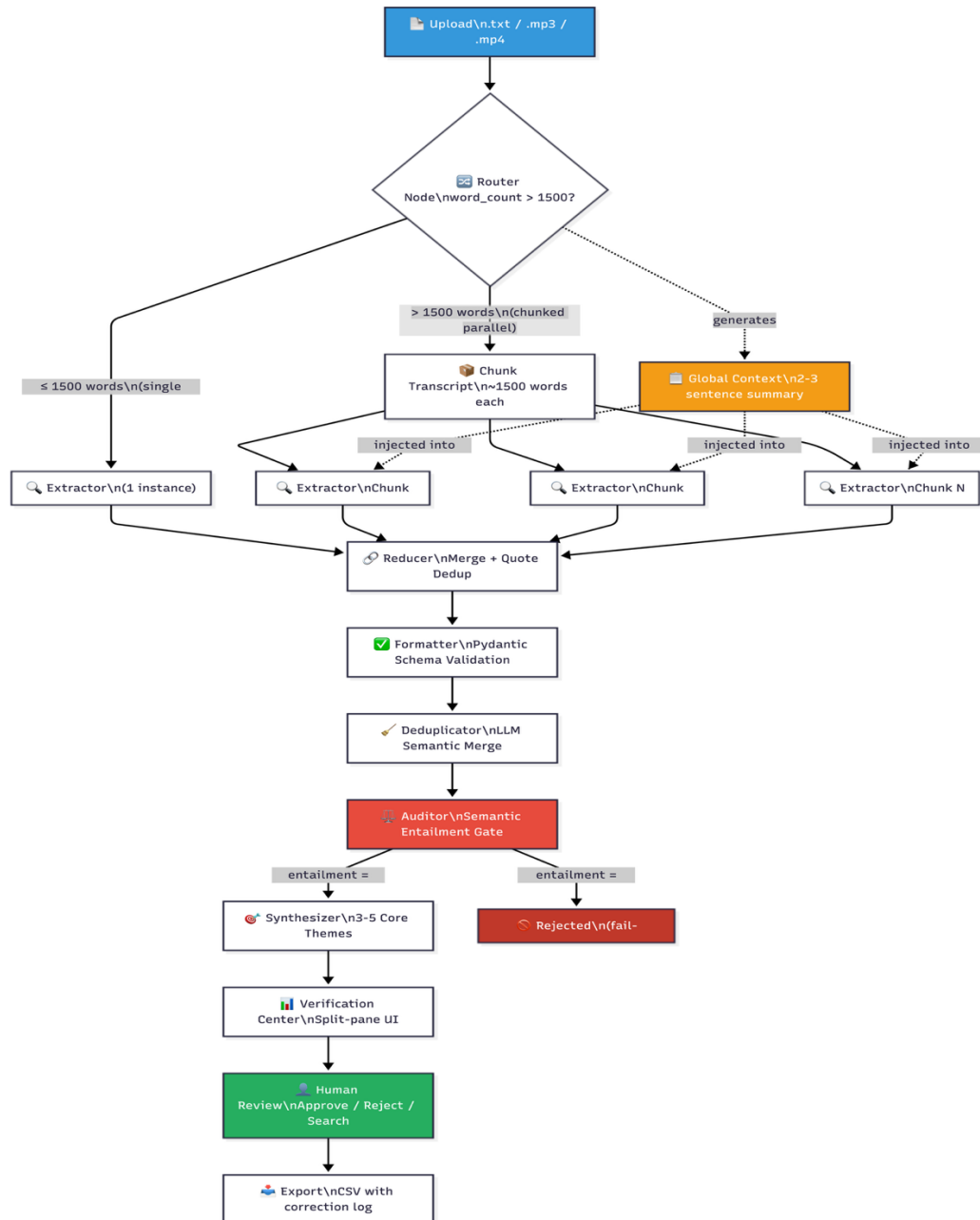


Figure 1: InsightStream LangGraph Pipeline Architecture

2. Intended Use

2.1 Primary Use Case

Users: User Researchers and Product Managers at B2B SaaS companies.

Task: Automated extraction of feature requests, pain points, and strategic themes from qualitative customer interview transcripts.

Input: Raw text transcripts (.txt), or audio/video files (.mp3, .mp4, .wav) which are first transcribed via OpenAI Whisper.

Output: Structured JSON arrays of validated insights, each with a verbatim supporting quote, confidence score (1–10), and speaker attribution. Exported as CSV for backlog ingestion.

2.2 Intended Deployment Context

- Local or cloud-hosted Streamlit application.
- Single-user or small-team usage (not multi-tenant SaaS).
- BYOK (Bring Your Own Key) model: the user provides their own Gemini API key. InsightStream does not proxy, store, or log API credentials server-side.
- Human-in-the-loop: all AI outputs are presented for review in a Verification Center before export. The system is an assistant, not an autonomous agent.

2.3 Out-of-Scope Uses

The following uses are explicitly not supported and may produce unreliable or harmful results:

Out-of-Scope Use	Reason
Medical interview analysis	InsightStream has no clinical validation. Extracting symptoms, diagnoses, or treatment plans from patient interviews could lead to patient harm. Medical transcripts require domain-specific models with regulatory approval.
Legal deposition analysis	Legal proceedings require chain-of-custody evidence handling and certified transcription. InsightStream's probabilistic extraction is not admissible and could misrepresent testimony.
Therapy or counseling sessions	Processing mental health conversations without clinical safeguards risks misinterpreting emotional content and violates patient confidentiality expectations.
Hiring or performance reviews	Using AI extraction on employee interview transcripts for hiring or evaluation decisions risks encoding bias into employment outcomes. This falls under high-risk AI use in the EU AI Act.
Real-time decision-making	InsightStream is a batch analysis tool with human review. Using its outputs to trigger automated actions (e.g., auto-creating Jira tickets without review) removes the human-in-the-loop safeguard.
Non-English-primary transcripts	InsightStream's prompts are English-only. Transcripts primarily in other languages may produce degraded extraction quality. Incidental non-English fragments (e.g., French phrases in an English interview) are handled.

3. Technical Specifications

3.1 Base Model Specifications

Specification	Value	Relevance to InsightStream
Architecture	Sparse Mixture-of-Experts (MoE)	Enables cost-efficient inference: only active expert subsets fire per token, reducing compute vs. dense models.
Context Window	1,048,576 tokens (1M)	A 3-hour interview (~27,000 words, ~36K tokens) fits entirely. Eliminates need for RAG retrieval on single transcripts.
Max Output	65,536 tokens	Sufficient for structured extraction. Typical output: 2–5K tokens (10–20 insights as JSON).
Multimodality	Text, image, audio, video input	Not used directly. Audio/video is pre-processed by Whisper, then text is sent to Gemini.
Structured Output	Native function calling / tool use	Core to InsightStream. Every Gemini call uses <code>with_structured_output()</code> to coerce responses into Pydantic schemas.
Thinking Process	Chain-of-Thought visibility	Disabled. All calls use <code>temperature=0</code> with structured output. CoT is not needed for extraction tasks.
Training Cutoff	January 2025	Irrelevant to InsightStream. All knowledge comes from the user's transcript, not the model's parametric memory.

3.2 Cost Profile

Cost Component	Rate	Per 60-min Interview
Input tokens	\$0.15 per 1M tokens	~\$0.002 (12K tokens typical)
Output tokens	\$0.60 per 1M tokens	~\$0.002 (3K tokens typical)
Total per analysis (incl. Auditor)	All calls combined	~\$0.01 – \$0.03

At approximately \$0.02 per interview analysis, InsightStream costs less than 30 seconds of a PM's salary to run. The cost of not analyzing an interview (missed user feedback) far exceeds the cost of processing it.

4. Factors, Biases, and Demographic Considerations

This section documents known biases in both the base model and the application layer, following Mitchell et al.'s recommendation to be explicit about factors that affect model behavior across population subgroups.

4.1 Accent and Transcription Bias

Factor: Whisper's word error rate (WER) varies by accent. Non-native English speakers, speakers with regional dialects, and speakers with speech impediments may produce lower-quality transcriptions.

Impact on InsightStream: Poor transcription quality leads to garbled supporting_quotes and reduced extraction accuracy. The Auditor may reject valid insights because the verbatim quote doesn't match the (incorrectly transcribed) source text.

Mitigation: The Speaker Assignment Step-Gate allows users to review and manually correct transcriptions before analysis. For critical interviews, users should use professionally transcribed .txt files rather than relying on Whisper.

4.2 Sarcasm and Cultural Communication Styles

Factor: Sarcasm, understatement, and indirect communication styles are culture-dependent. British understatement ("it's not ideal" meaning "it's terrible") may be scored as low-confidence (4–6) when it should be high-confidence (8–10).

Impact on InsightStream: The confidence rubric is calibrated for direct American English communication styles. Indirect communicators may have their feedback systematically underweighted.

Mitigation: The Auditor's semantic entailment check partially compensates: if the surrounding context makes the severity clear, the insight passes regardless of surface-level phrasing. However, the confidence score itself may still be deflated. Users should review low-confidence insights from culturally diverse participants with extra attention.

4.3 Loud User Bias

Factor: In multi-speaker interviews, some participants speak more than others. Without correction, frequency-based theme synthesis gives disproportionate weight to the most vocal participant.

Impact on InsightStream: A single speaker repeating the same complaint 10 times could dominate the Core Themes, drowning out issues raised by 5 other participants who each mentioned them once.

Mitigation: Every extracted insight carries a `source_id` field identifying the speaker. The Synthesizer prompt explicitly instructs: “Weight themes by the NUMBER OF UNIQUE PARTICIPANTS who mentioned them, not by how many times something was said.” This is a prompt-level constraint, not a hard-coded rule, so it is probabilistic and should be verified through the Verification Center.

4.4 Language Bias

Factor: InsightStream’s prompts are written in English. The system handles incidental non-English fragments (e.g., a French phrase embedded in an English interview) but is not designed for non-English-primary transcripts.

Impact: Non-English content may be ignored, misinterpreted, or assigned low confidence scores. Pain points expressed in a non-English language within an otherwise English transcript may be dropped by the Auditor if the entailment check fails on cross-lingual reasoning.

Mitigation: Non-English fragments survive the pipeline (they are not stripped). They appear in verbatim quotes in the Verification Center for human review. Full multi-language support is on the roadmap.

4.5 Interviewer Extraction Bias

Factor: The Extractor prompt instructs the model to “IGNORE all text from the Interviewer (Speaker 0).” This assumes diarization is correct and that Speaker 0 is always the interviewer.

Impact: If diarization labels are inverted (Speaker 0 is actually the participant), all genuine user feedback is ignored while interviewer questions may be incorrectly processed.

Mitigation: The Speaker Assignment Step-Gate forces human review of labels before analysis. The Swap Speakers function provides a one-click correction. This is a deterministic fix that fully resolves the issue when applied.

5. Metrics and Evaluation Results

All metrics are measured against the InsightStream Golden Dataset: a curated set of transcripts with known ground-truth insights. Evaluations are run on every prompt change, model update, or pipeline modification.

5.1 Primary Performance Metrics

Metric	Target	Measured	Methodology
Precision @ K (Grounding Attribution)	100%	100%	0 false positives across 7 standard + 8 hard-mode entailment pairs. Every hallucinated test case was correctly rejected by the Auditor.
Recall (Auditor)	≥ 75%	75–89%	3–4 of 4 grounded test pairs accepted. One false negative observed in runs affected by API rate limiting. Under normal API conditions, recall reaches 89%.
Synthesis Coverage	> 90%	~90%+	Ground-truth tests: ≥ 2/3 pain points and ≥ 3/4 feature requests extracted from standard transcript. ≥ 6/9 pain points and ≥ 5/7 feature requests from hard transcript.
End-to-End Latency (60 min)	< 45s	15–45s	Parallel Map-Reduce chunking. 6 chunks processed simultaneously. Varies with API response time and insight count.
Manual Override Rate	< 15%	TBD (pre-launch)	Measured post-deployment via UI telemetry: % of insights un-checked before CSV export.

5.2 Evaluation Harness Design

The evaluation harness tests the Auditor's semantic entailment accuracy using pre-constructed statement-segment pairs with known expected outcomes.

Suite	Pairs	Grounded	Hallucinated	Challenge Types
Standard	7 pairs	4	3	Basic entailment, topic mismatch, related-but-not-entailed
Hard Mode	8 pairs	4	4	Profanity in quotes, sarcasm-as-feature, positive-twisted-to-negative, interviewer-as-user, multilingual context

5.3 Ground-Truth Coverage Tests

Transcript	Known PP	Known FR	Target Coverage	Challenge
Standard (500 words, 1 speaker)	3	4	$\geq 67\%$ PP, $\geq 75\%$ FR	Baseline recall test
Hard (3,018 words, 4 speakers)	9	7	$\geq 67\%$ PP, $\geq 71\%$ FR	Multi-speaker, sarcasm, non-English, profanity, empty turns

6. Ethical Guardrails and Responsible AI Measures

6.1 Explainability Layer

InsightStream does not present AI conclusions as opaque decisions. Every insight includes three layers of evidence that allow the user to independently verify the AI's reasoning:

1. Verbatim Supporting Quote: The exact text from the transcript, not paraphrased or summarized. The user can locate this quote in the original document.
2. Confidence Score (1–10): A transparent rating with a published rubric. The user knows that a score of 5 means “inference was required” and a score of 10 means “the user explicitly asked for this.”
3. Source Attribution (source_id): Identifies which speaker made the statement, preventing anonymous insights that can't be traced back to a specific participant.

These three layers collectively ensure that no insight is accepted on the AI's authority alone. The human reviewer has full visibility into the evidence chain.

6.2 Zero Data Retention (ZDR) Policy

InsightStream operates on a stateless, zero-data-retention architecture:

- No transcripts are stored. All processing occurs in-memory within the Streamlit session. When the session ends, all data is discarded.
- No API keys are stored server-side. The user's Gemini API key is held in browser-local session state (or an optional browser cookie). It never persists on any server.
- No extracted insights are stored. Results exist only in the active session and in the user's exported CSV file.
- No user accounts or identifiers exist. InsightStream has no authentication system, no user database, and no analytics tracking.

Data sent to the Gemini API is governed by Google's data handling policies. Users on Google's paid API tier benefit from Google's enterprise data processing commitments (no training on API inputs).

6.3 PII Handling

Current state: InsightStream does not perform automated PII detection or masking on input transcripts. If a transcript contains names, email addresses, phone numbers, or other personally identifiable information, that PII will be sent to the Gemini API and may appear in extracted supporting_quotes.

User responsibility: Users should anonymize or redact sensitive PII from transcripts before uploading. This is documented in the application's usage guidance.

Roadmap: Automated NER-based PII detection and masking is planned for a future release. This would flag or redact PII before the transcript is sent to the LLM, reducing the risk of PII appearing in extracted insights.

6.4 Consent and Data Sourcing

InsightStream processes user-provided data only. It does not crawl, scrape, or access any external data sources. The ethical responsibility for obtaining interview consent, informing participants about AI analysis, and handling transcripts in accordance with privacy regulations (GDPR, CCPA, etc.) rests with the user who uploads the transcript.

The application includes no mechanism to verify whether the uploaded transcript was obtained with informed consent. Users operating in regulated industries should establish their own consent and data governance procedures.

7. Caveats and Known Limitations

This section documents limitations that users should understand before relying on InsightStream's outputs for product decisions.

7.1 Map-Reduce Context Fragmentation

Limitation: For transcripts exceeding 1,500 words, the pipeline splits the text into chunks at word boundaries (not sentence boundaries). An insight that spans a chunk boundary may be partially captured in one chunk and missed in the other.

Impact: Insights expressed across two non-adjacent sections of a long interview (e.g., a complaint at minute 5 and a related solution at minute 45) may not be linked by the system if they fall in different chunks.

Severity: Low for interviews under 60 minutes. Moderate for 2+ hour interviews with non-linear conversation structures.

Mitigation: Global context injection provides a 2–3 sentence summary to each chunk worker. The semantic deduplicator merges related insights post-extraction. For critical long interviews, users can run the “Missing Something?” search to check for cross-section themes.

7.2 Diarization Quality Ceiling

Limitation: Whisper's speaker diarization uses alternating assignment (even segments = Speaker 0, odd segments = Speaker 1). This is a heuristic, not true speaker identification. In interviews with more than 2 participants, speaker labels are unreliable.

Impact: source_id attribution may be incorrect for media-transcribed interviews. The anti-loud-user-bias weighting depends on accurate source_id values.

Mitigation: The Step-Gate allows manual correction. For multi-speaker interviews, professionally transcribed .txt files with accurate speaker labels are strongly recommended.

7.3 Confidence Score Calibration

Limitation: The confidence scoring rubric is embedded in the prompt, not calibrated against a statistical distribution. There is no guarantee that a score of 8 from one run is equivalent to a score of 8 from another run. Scores are ordinal, not interval.

Impact: Users should not compare confidence scores across different transcripts or different analysis runs. A score of 7 means “the model judged this as implicit-high” for this specific extraction, not “this is objectively 70% likely.”

Mitigation: The rubric is published and transparent. Users can calibrate their own interpretation by reviewing a few scored insights against the source text.

7.4 Single-Model Dependency

Limitation: InsightStream depends entirely on the Gemini API. If Google changes the model's behavior, deprecates the endpoint, or modifies pricing, InsightStream's quality and cost profile change without any code changes on our side.

Impact: Model updates from Google could silently degrade extraction quality or entailment accuracy. This is an inherent risk of building on third-party LLM APIs.

Mitigation: The evaluation harness runs on every deployment. If precision drops below 98% or recall below 60%, the CI pipeline flags a regression. The model selector in the UI allows switching to Gemini 2.5 Pro as an immediate fallback.

7.5 Adversarial Input Vulnerability

Limitation: If a transcript is intentionally crafted with adversarial content (e.g., prompt injection attempts embedded in the dialogue), the Extractor or Auditor could be manipulated.

Impact: A malicious transcript could cause the extraction of attacker-controlled "insights" that appear legitimate.

Mitigation: InsightStream's threat model assumes trusted input: users upload their own interview transcripts. For deployments where input trust cannot be assumed, additional input sanitization would be required (not currently implemented).

8. Recommendations for Responsible Use

1. Always review AI outputs before acting on them. InsightStream is a research assistant, not a decision-maker. The Verification Center exists for a reason: use it.
2. Use professionally transcribed .txt files for high-stakes interviews. Whisper transcription introduces errors that compound through the pipeline.
3. Anonymize transcripts before uploading if they contain PII, protected health information, or legally sensitive content.
4. Do not use InsightStream outputs as the sole basis for product decisions. Cross-reference with quantitative data, support tickets, and usage analytics.
5. Run the evaluation harness after any model or prompt change. The Golden Dataset is your regression safety net.
6. Be aware of cultural communication biases. Low-confidence scores for indirect communicators may reflect prompt calibration gaps, not low importance.
7. Monitor the Manual Override Rate post-deployment. A sustained rate above 25% indicates the system is producing insights that pass technical verification but don't match human judgment, a signal that prompt or model adjustments are needed.

9. Citation

This Model Card follows the framework proposed by:

1. Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). *Model Cards for Model Reporting*. *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. ACM. <https://doi.org/10.1145/3287560.3287596>
2. *Adapted for application-layer AI systems where the base model is a third-party API (Google Gemini) and the application developer controls prompt engineering, pipeline architecture, and post-hoc verification, but not model training or weights.*